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SUB QUERIES

1 What is a sub query?

A Subquery is a query written inside another SQL query

- Inner Query: The query inside another query
- Outer Query: The main query that uses the inner query's result

→ First SQL finds an answer to a smaller questions
→ Then it uses that answer to solve the main question

Syntax:
```
SELECT column_name
FROM table_name
WHERE column_name = (
    SELECT column_name
    FROM table_name
    WHERE condition
);
```

The inner query is placed inside parentheses()

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2 When do we use Subqueries?

We use subqueries when we need to fetch data from one table based on information obtained from a query on another table or from the same table

Ex: find the employee earning the highest Salary

Suppose we have an employee table

emp_name	Salary
Ravi	20,000
Priya	30,000
Arun	45,000
Meena	25,000

```
SELECT emp_name, Salary
FROM employee
WHERE Salary = (
   SELECT MAX (salary)
    FROM employee
);
```

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3 How does a subquery execute?

Step 1: Inner query
Find the required value
Ex: Highest salary = 50000

↓

Step 2: outer query
use the value to retrieve matching row
find the employee earning 50,000

4 Types of subqueries

i Single row subquery
ii Multi row subquery
iii Correlated subquery

i Single row subquery
A subquery that returns only one row (one result row)
Use the (=) operator when the inner query returns a single value

Ex: Employee table

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emp_name	department	Salary
Ravi	IT	30,000
Priya	HR	25,000
Arun	IT	45,000
Meena	HR	35,000

Find the employee who earns the highest Salary

```
SELECT emp_name, salary
FROM employee
WHERE Salary = (
  SELECT MAX (salary)
    FROM employee
);
```

Inner query : MAX(salary) returns 45000
outer query : finds the employee whose Salary equals 45000

O/P :

emp_name	Salary
Arun	45,000

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ii Multi row subquery

A subquery that returns more than one row

Use the (IN) operator to match a value against multiple results

find employees who work in departments that have employees earning more than ₹40,000

```
SELECT emp.name, department, salary
 FROM employees
WHERE department IN (
    SELECT department
    FROM employees
    WHERE salary >40000
);
```

→ Inner query finds the departments of employees earning more than ₹40,000

→ It returns IT, because Arun earns ₹45,000

→ Outer query returns all employees whose department is IT

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o/p : emp_name | department | Salary
Ravi | IT | 30,000
Arun | IT | 45,000

iii Correlated subquery

A subquery that refers to a column or value from the outer query
Unlike a simple subquery, it depends on the current row being considered by the outer query

Ex: find employees earning more than their departments average salary

```
SELECT emp_name, department, salary
FROM employee e1
WHERE salary > (
    SELECT AVG (salary)
    FROM employee e2
    WHERE e1.department = e2.department
);
```

Department	Salaries	Average
IT	30,000 ; 45,000	37,500
HR	25,000 ; 35,000	30,000

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Now compare each employee's salary with their department average

o/p : emp-name	department	Salary
Arun	IT	45,000
Meena	HR	35,000

→ Arun earns more than the IT average and
→ Meena earns more than the HR average

Type	Inner query returns	Common operator or feature
* Single-row	one row	=
* Multi-row	multiple rows	IN
* correlated	Depends on the outer row	References outer query

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